YouMentor AI

AI-Powered Multilingual Learning Assistant

Project Domain: Artificial Intelligence | EdTech | Multilingual Accessibility
Institution: Maturi Venkata Subba Rao Engineering College (MVSR)

1. Project Summary

YouMentor AI is an AI-powered Chrome extension that transforms YouTube videos into interactive, multilingual learning experiences. It combines speech recognition, translation, AI voice generation, and intelligent study tools to help students understand, summarize, and revise educational content in their preferred language.

2. Problem Statement

Students often face language barriers while learning through online videos. Understanding unfamiliar concepts, accessing content in regional languages, and preparing revision notes from lengthy videos can be challenging and time-consuming.

3. Proposed Solution

YouMentor AI provides a unified learning assistant that converts video content into understandable subtitles, translated speech, summaries, explanations, quizzes, and flashcards.

4. Key Features

- **Live Subtitles:** Real-time speech recognition and multilingual subtitles using Gnani Prisma STT.
- **AI Voice:** Listen to translated content in a selected language using speech synthesis.
- **Caption-Based Voice:** Uses original YouTube captions and timestamps to generate translated speech synchronized with video playback.
- **AI Summarization:** Generates video overviews, key points, important concepts, and takeaways.
- **AI Explanation:** Simplifies difficult topics for better understanding.
- **Quizzes & Flashcards:** Creates revision materials from learning content.
- **AI Notebook:** Saves and organizes important notes for future reference.

5. Technology Stack

Frontend: HTML, CSS, JavaScript, Chrome Extension (Manifest V3)
Backend: Python, Flask, REST APIs, WebSockets
AI & Speech: Groq LLM, Gnani Prisma STT, Gnani TTS
Other Tools: YouTube Transcript API, Chrome Storage

6. Innovation & Impact

YouMentor AI integrates multilingual speech technologies with AI-powered study tools in one platform. Its live speech recognition supports ten Indian languages: Bengali, English, Gujarati, Hindi, Kannada, Malayalam, Marathi, Punjabi, Tamil, and Telugu.

It aims to make digital education more inclusive by reducing language barriers, improving comprehension, and simplifying revision.

Project Vision

Learn without language barriers. Understand without limits.